

Project Design Phase – EduGenie

1. System Architecture

- EduGenie is designed as a web-based AI learning assistant.
- The system connects the user interface with the backend and Google Gemini AI.
- User questions are sent to the AI system and the generated response is displayed to the student.

2. User Interface Design

- Simple and user-friendly interface.
- Chat-based screen for asking questions.
- Clear input box and response area.
- Easy navigation for students.

3. AI Module Design

- Google Gemini is used to understand student queries.
- The AI generates relevant and simple explanations.
- The system can support summaries, study notes, quizzes, and practice questions.

4. Backend Design

- Handles user requests and communication with the AI model.
- Processes inputs and returns AI-generated responses.
- Provides the required connection between the frontend and AI services.

5. Database / Storage Design

- User-related information can be stored securely when required.
- Study history or generated content can be maintained if the application supports these features.

6. Security Design

- User data should be protected.
- API keys and sensitive credentials should not be exposed in the frontend.
- Access to stored information should be controlled.

7. Overall Workflow

Student → EduGenie Interface → Backend → Google Gemini AI → Response → Student

8. Expected Outcome

The project design provides a clear structure for developing EduGenie as a simple, interactive, secure, and AI-powered learning assistant.

